CS177 Bioinformatics: Software Development and Applications

Lecture 23: AI Virtual Cell (AIVC)

Jie Zheng (郑杰)
PhD, Associate Professor
School of Information Science and Technology (SIST), ShanghaiTech University

Spring, 2025

Learning objectives

- At the end of this lecture, you should be able to:
  - Understand the basics of whole-cell modeling (aims, methodology, and challenges)
  - Describe how to model cell cycle control system
  - Explain the key difference between whole-cell modeling and AI virtual cell (AIVC)
  - Define and explain the concept of universal representation (UR) at multiple scales in molecular and cell biology
  - Describe the virtual instruments (VIs), including the manipulators and decoders
  - Give a few examples of applying AIVC, e.g. drug discovery, cell engineering, cancer medicine, scientific research

Whole-Cell Modeling

Aims of whole-cell modeling

- Whole-cell models aim to
  - Represent all of the chemical reactions in a cell and all of the physical processes that influence their rates
  - Predict the behavioral trajectories of single cells over their life cycles, with each simulation representing a different cell within a heterogeneous clonal population

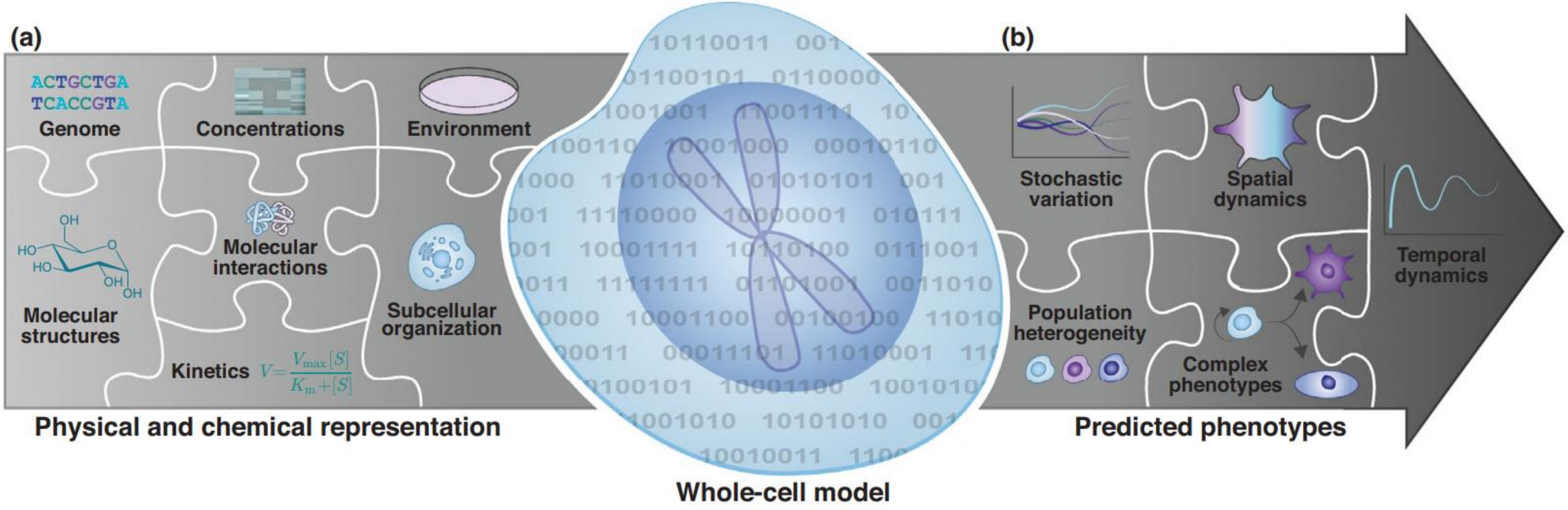


Fig. 1 of Goldberg et al. (2018) Current Opinion in Biotechnology, **51**:97–102

Whole-cell modeling methodology

- Recent advances in data aggregation, model design, model representation, and simulation are also rapidly making whole-cell modeling feasible. Emerging methods and tools include:
  - Data aggregation and organization
  - Scalable model design
  - Model languages
  - Simulation, calibration and verification
  - Simulation results analysis

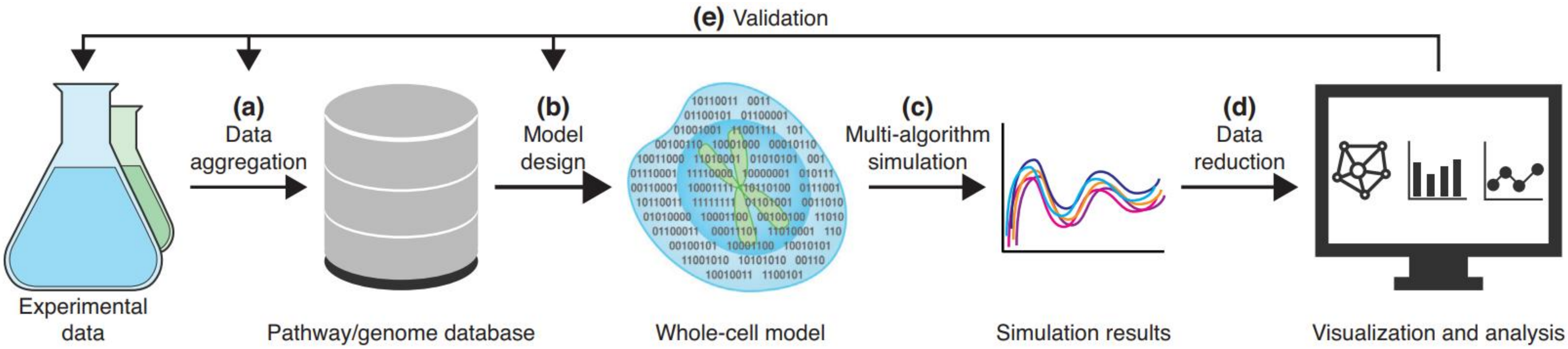


Fig. 2 of Goldberg et al. (2018) Current Opinion in Biotechnology, **51**:97−102

Technical challenges

- A recent survey of the bio-modeling community (Szigeti et al. 2018) shows the following bottlenecks to whole-cell modeling:
  - Inadequate experimental methods and data repositories
  - Incompatibility among published models
  - Inadequate software tools for designing, calibrating and validating models
  - Lack of adequate formats for representing models

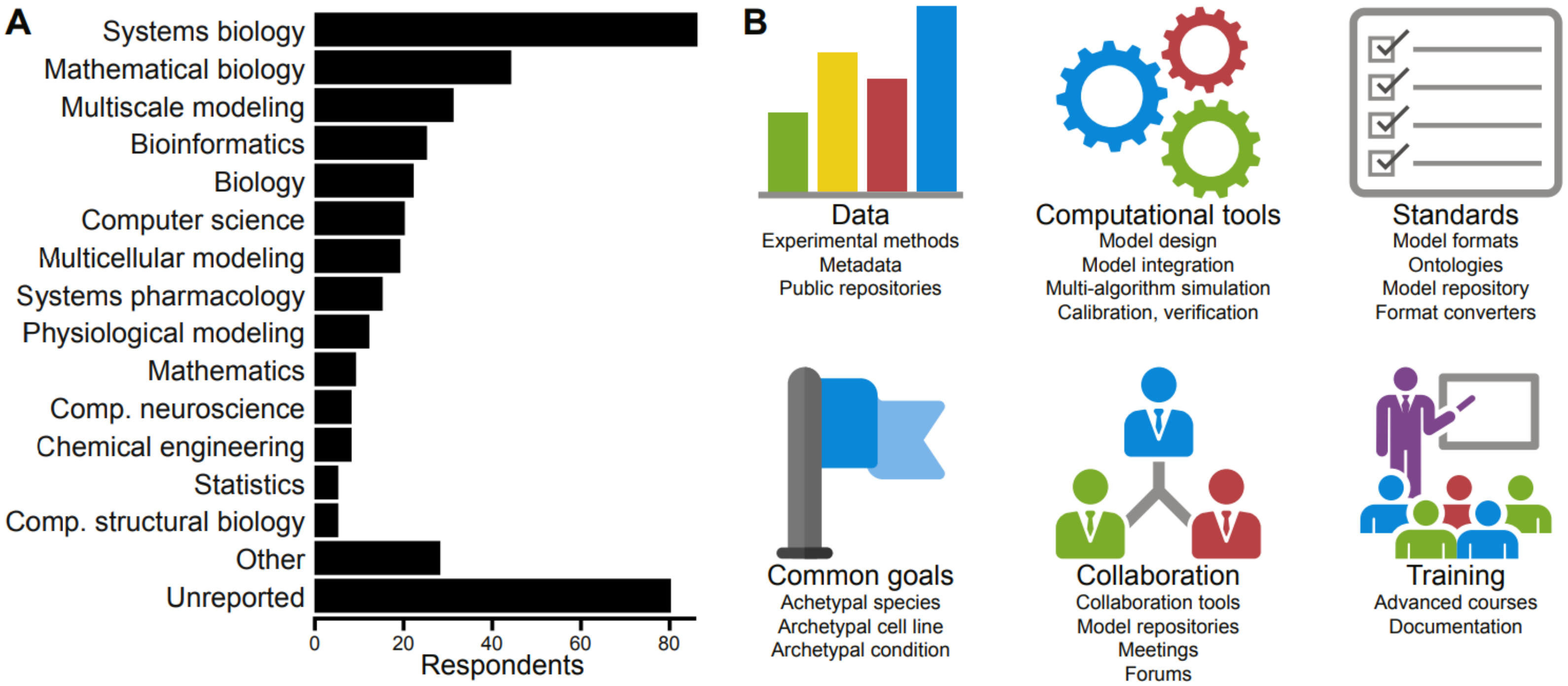
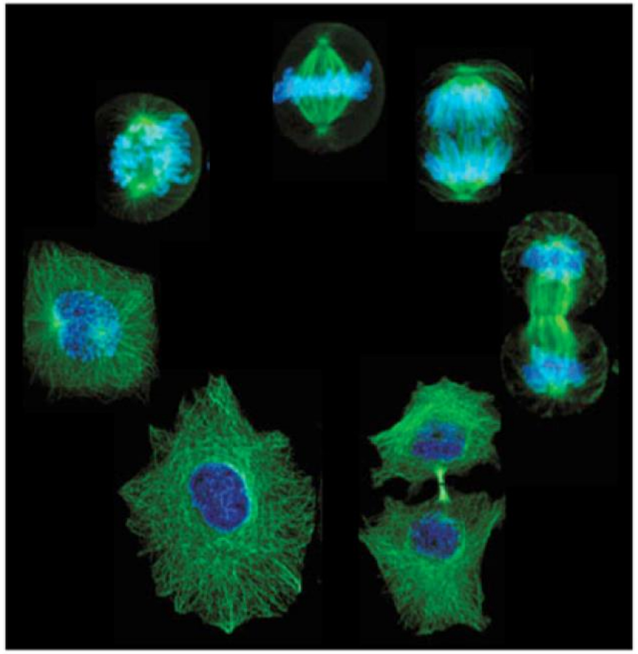


Fig. 2 of Szigeti et al. (2018) Current Opinion in Systems Biology, **7**:8−15

In Silico Simulation of Cell Cycle Control

The cell cycle

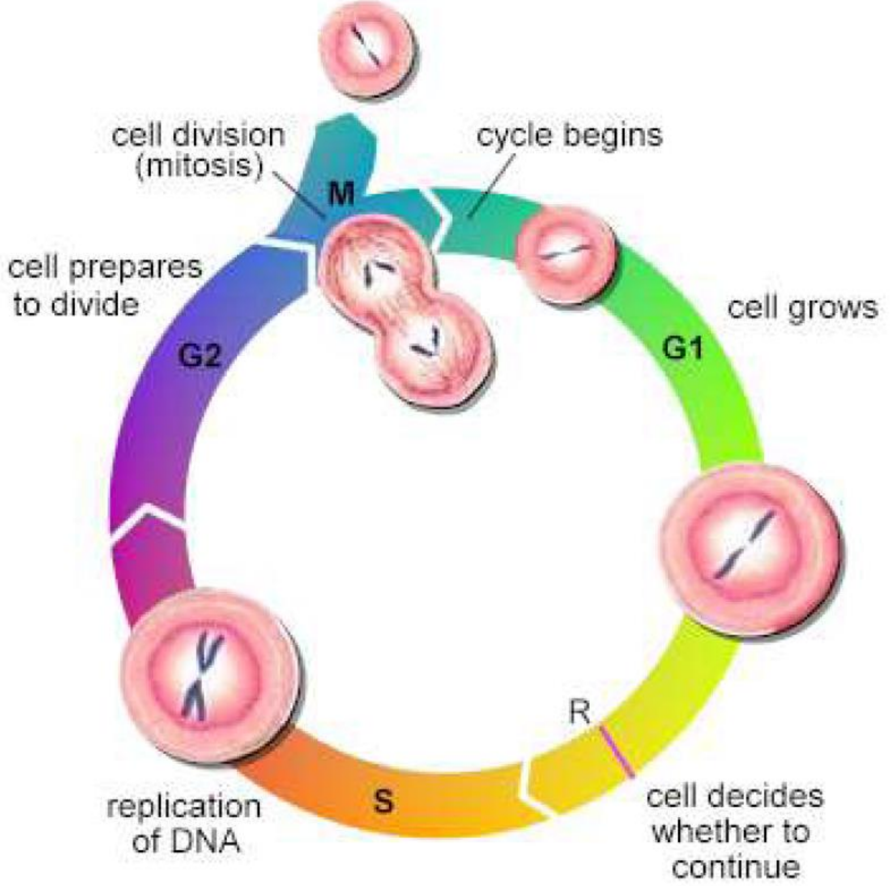
- Among the cellular processes, the cell division cycle is one of the most fascinating and well studied, due to
  - The fine regulation mechanisms used for the synchronization and proof checking of the process
  - The relevance for understanding the occurrence of many diseases, such as cancer



The cell cycle (cont'd)

- Each cell reproduces by performing an orderly sequence of events
- The classical model of cell cycle comprises 4 sequential phases:

- **S (synthesis) phase**: The DNA is replicated to produce two identical copies
- **G2 (gap) phase**: New proteins are synthesized and cell approximately doubles in size
- **M (mitosis) phase**: Matching chromosomes are pulled to opposite poles of the cell and cytokinesis pinches the cell in half
- **G1 (gap) phase**: Each daughter cell replicates again after a gap period, or enters a quiescent phase (G0)



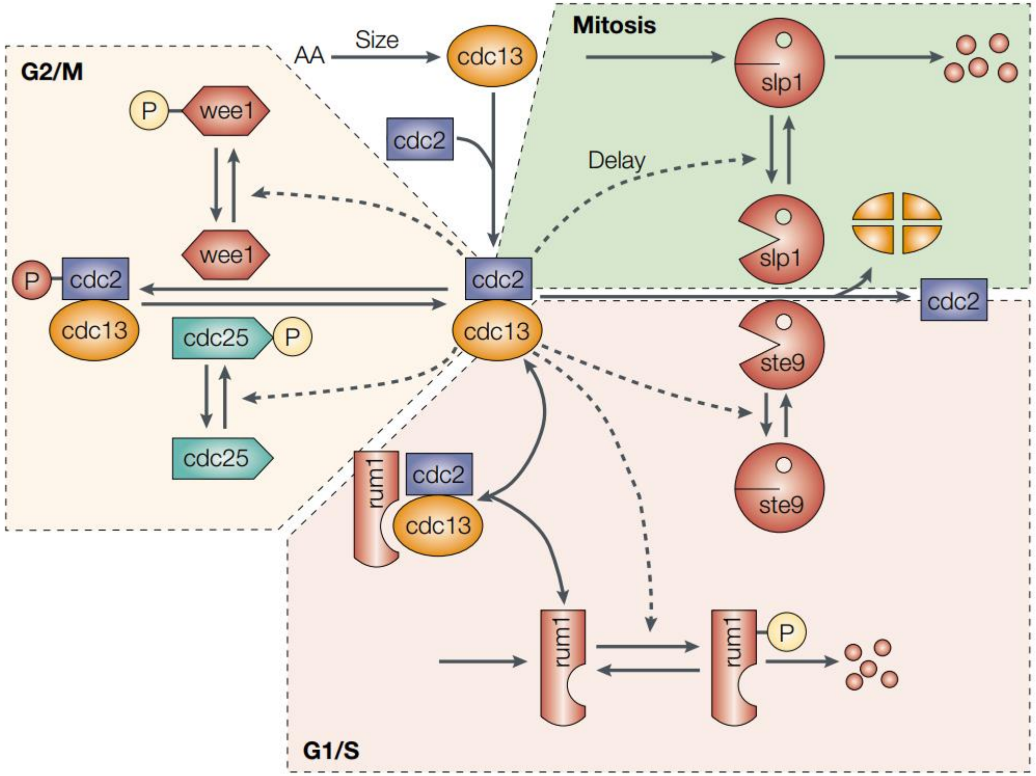
立志 成才 报国 裕民

Cell cycle control system

- In the sequel, we will follow the presentation given in

  Tyson, Chen, Novak, Network Dynamics and Cell Physiology, Nature Reviews Molecular Cell Biology, 2:908-916, 2001

  - The scheme represents a wiring diagram of the basic mechanisms involved in the regulation of the cell cycle
  - Each node represents a protein (or complex of proteins)
  - Solid arrows represent the biochemical process that produce, degrade, activate or inactivate the attached proteins
  - Dashed arrows represent the regulatory influence of one protein on a biochemical process



The cell-cycle control system in fission yeast

Cell cycle in mammalian cells

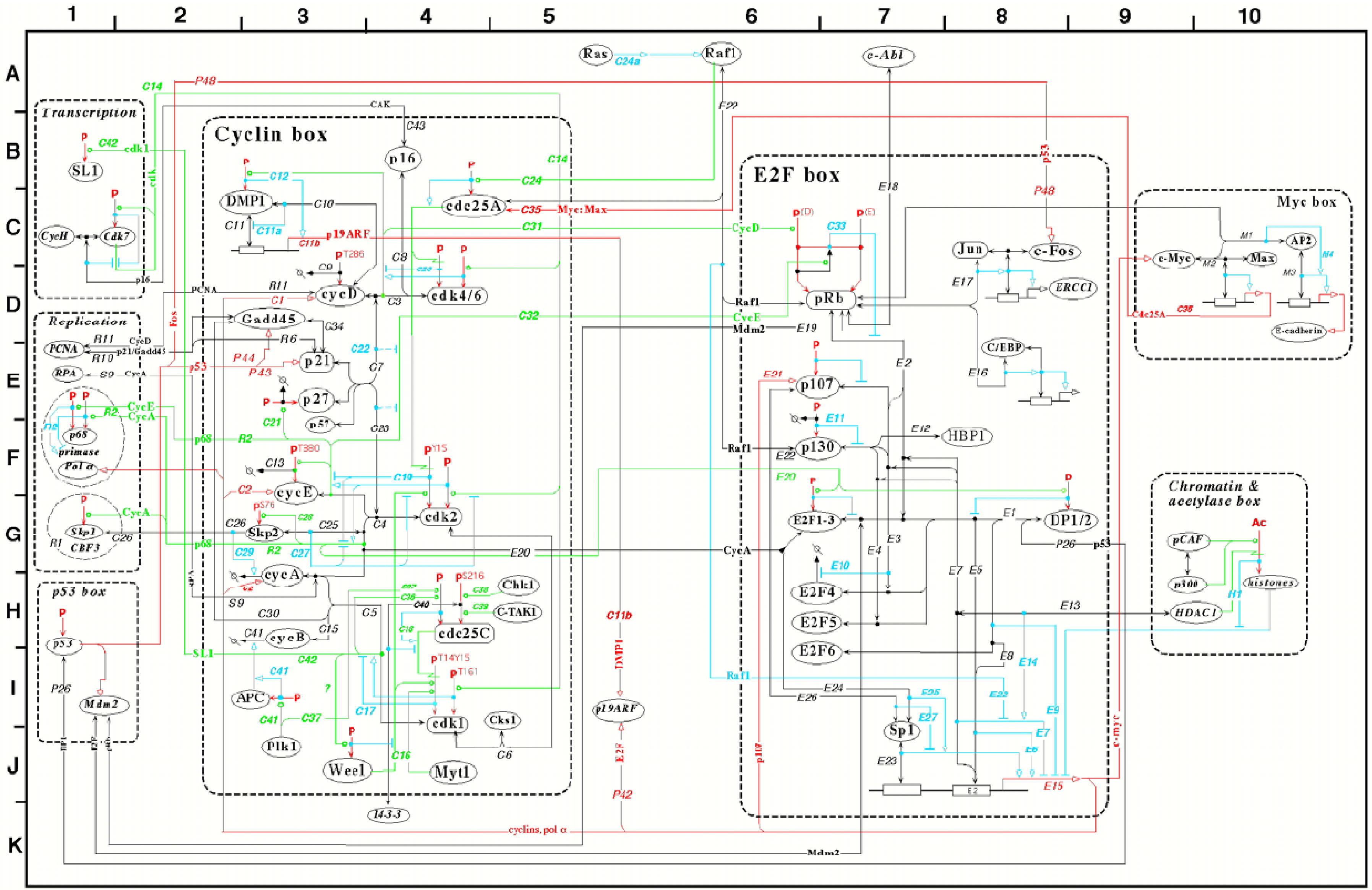


Figure 6A: The Cyclin - E2F cell cycle control system (version 3a - June 8, 1999)

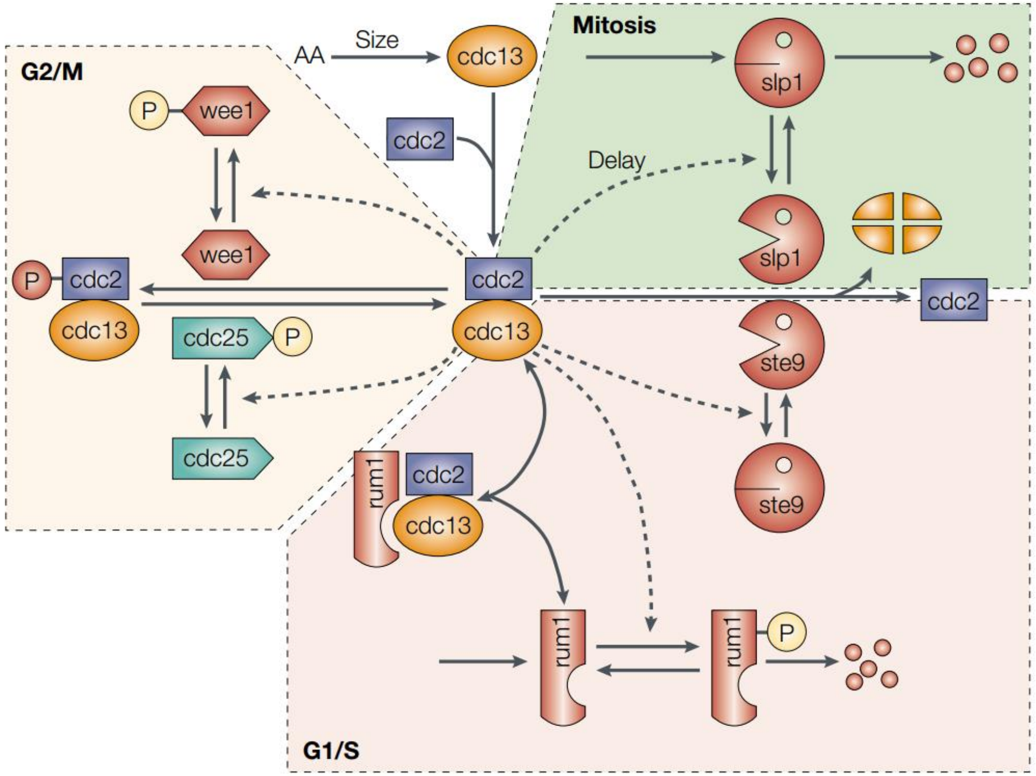
立志 成才 报国 裕民

Model derivation

- The ODE model can be derived as:

$$d/dt\,[\mathrm{cdc2\text{-}cdc13}] = k_1 \cdot \mathrm{size} - k_2[\mathrm{cdc2\text{-}cdc13}] - k_3[\mathrm{rum1}][\mathrm{cdc2\text{-}cdc13}] + k_4[\mathrm{rum1\text{-}cdc2\text{-}cdc13}] - k_5[\mathrm{cdc2\text{-}cdc13}] + k_6[\mathrm{cdc2P\text{-}cdc13}]$$

- The kinetic parameters are, in turn, determined by the activity of the other species
- For instance, the parameter k_2 quantifies the degradation rate of cdc13, which depends on the activities of slp1 and ste9



Model equations

$$\frac{d[\mathrm{Cdc13}_T]}{dt}=k_1M-(k_2'+k_2''[\mathrm{Ste9}]+k_2'''[\mathrm{Slp1}])[\mathrm{Cdc13}_T], \quad (1)$$

$$\frac{d[\mathrm{preMPF}]}{dt}=k_{\mathrm{wee}}([\mathrm{Cdc13}_T]-[\mathrm{preMPF}])-k_{25}[\mathrm{preMPF}]-(k_2' + k_2''[\mathrm{Ste9}]+k_2'''[\mathrm{Slp1}])[\mathrm{preMPF}], \quad (2)$$

$$\frac{d[\mathrm{Ste9}]}{dt}=(k_3'+k_3''[\mathrm{Slp1}])\frac{1-[\mathrm{Ste9}]}{J_3+1-[\mathrm{Ste9}]}-(k_4'[\mathrm{SK}] + k_4[\mathrm{MPF}])\frac{[\mathrm{Ste9}]}{J_4+[\mathrm{Ste9}]}, \quad (3)$$

$$\frac{d[\mathrm{Slp1}_T]}{dt}=k_5'+k_5''\frac{[\mathrm{MPF}]^4}{J_5^4+[\mathrm{MPF}]^4}-k_6[\mathrm{Slp1}_T], \quad (4)$$

$$\frac{d[\mathrm{Slp1}]}{dt}=k_7[\mathrm{IEP}]\frac{[\mathrm{Slp1}_T]-[\mathrm{Slp1}]}{J_7+[\mathrm{Slp1}_T]-[\mathrm{Slp1}]} - k_8\frac{[\mathrm{Slp1}]}{J_8+[\mathrm{Slp1}]}-k_6[\mathrm{Slp1}], \quad (5)$$

$$k_{\mathrm{wee}}=k_{\mathrm{wee}}'+(k_{\mathrm{wee}}''-k_{\mathrm{wee}}')G(V_{\mathrm{awee}},V_{\mathrm{iwee}}[\mathrm{MPF}],J_{\mathrm{awee}},J_{\mathrm{iwee}}),$$

$$k_{25}=k_{25}'+(k_{25}''-k_{25}')G(V_{a25}[\mathrm{MPF}],V_{i25},J_{a25},J_{i25}),$$

$$\Sigma=[\mathrm{Cdc13}_T]+[\mathrm{Rum1}_T]+K_{\mathrm{diss}},$$

$$G(a,b,c,d)=\frac{2ad}{b-a+bc+ad+\sqrt{(b-a+bc+ad)^2-4ad(b-a)}}.$$

$$\frac{d[\mathrm{IEP}]}{dt}=k_9[\mathrm{MPF}]\frac{1-[\mathrm{IEP}]}{J_9+1-[\mathrm{IEP}]}-k_{10}\frac{[\mathrm{IEP}]}{J_{10}+[\mathrm{IEP}]}, \quad (6)$$

$$\frac{d[\mathrm{Rum1}_T]}{dt}=k_{11}-(k_{12}+k_{12}'[\mathrm{SK}]+k_{12}''[\mathrm{MPF}])[\mathrm{Rum1}_T], \quad (7)$$

$$\frac{d[\mathrm{SK}]}{dt}=k_{13}[\mathrm{TF}]-k_{14}[\mathrm{SK}], \quad (8)$$

$$\frac{dM}{dt}=\mu M, \quad (9)$$

$$[\mathrm{Trimer}]=\frac{2[\mathrm{Cdc13}_T][\mathrm{Rum1}_T]}{\Sigma+\sqrt{\Sigma^2-4[\mathrm{Cdc13}_T][\mathrm{Rum1}_T]}}, \quad (10)$$

$$[\mathrm{MPF}]=\frac{([\mathrm{Cdc13}_T]-[\mathrm{preMPF}])([\mathrm{Cdc13}_T]-[\mathrm{Trimer}])}{[\mathrm{Cdc13}_T]}, \quad (11)$$

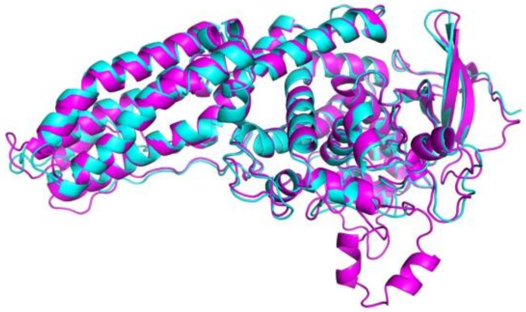
$$[\mathrm{TF}]=G(k_{15}M,k_{16}'+k_{16}''[\mathrm{MPF}],J_{15},J_{16}), \quad (12)$$

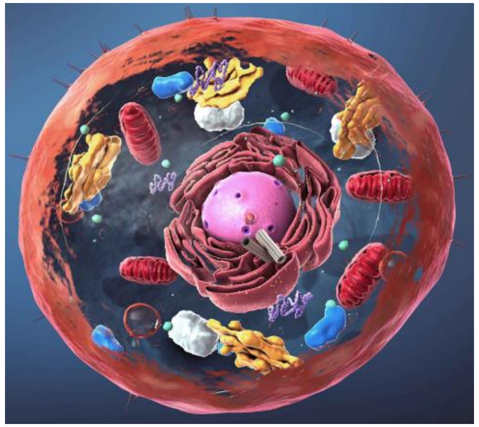
- 9 ODEs
- 3 algebraic equations
- ~30 parameters

Novak et al. Chaos 11(1):277-286, 2001

AI Virtual Cell (AIVC)

AI for Life Science: from protein to cell





AlphaFold

AI Virtual Cell (AIVC)

Structural Biology

Systems Biology

From Whole-Cell Modeling to AIVC

上海科技大学
ShanghaiTech University

Cell 2012 (Stanford)

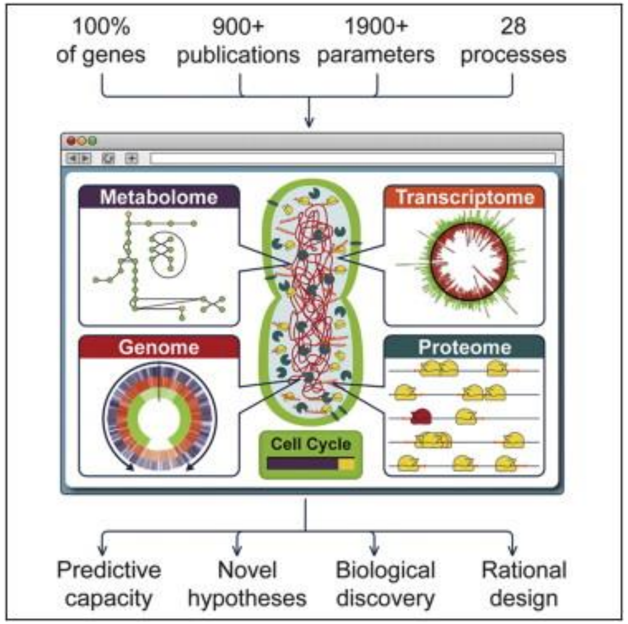
Cell 2024 (Stanford)

Theory

Cell

A Whole-Cell Computational Model Predicts Phenotype from Genotype

Jonathan R. Karr,[1,4] Jayodita C. Sanghvi,[2,4] Derek N. Macklin,[2] Miriam V. Gutschow,[2] Jared M. Jacobs,[2] Benjamin Bolival, Jr.,[2] Nacyra Assad-Garcia,[3] John I. Glass,[3] and Markus W. Covert[2,*]
[1]Graduate Program in Biophysics
[2]Department of Bioengineering
Stanford University, Stanford, CA 94305, USA
[3]J. Craig Venter Institute, Rockville, MD 20850, USA
[4]These authors contributed equally to this work
*Correspondence: mcovert@stanford.edu
http://dx.doi.org/10.1016/j.cell.2012.05.044



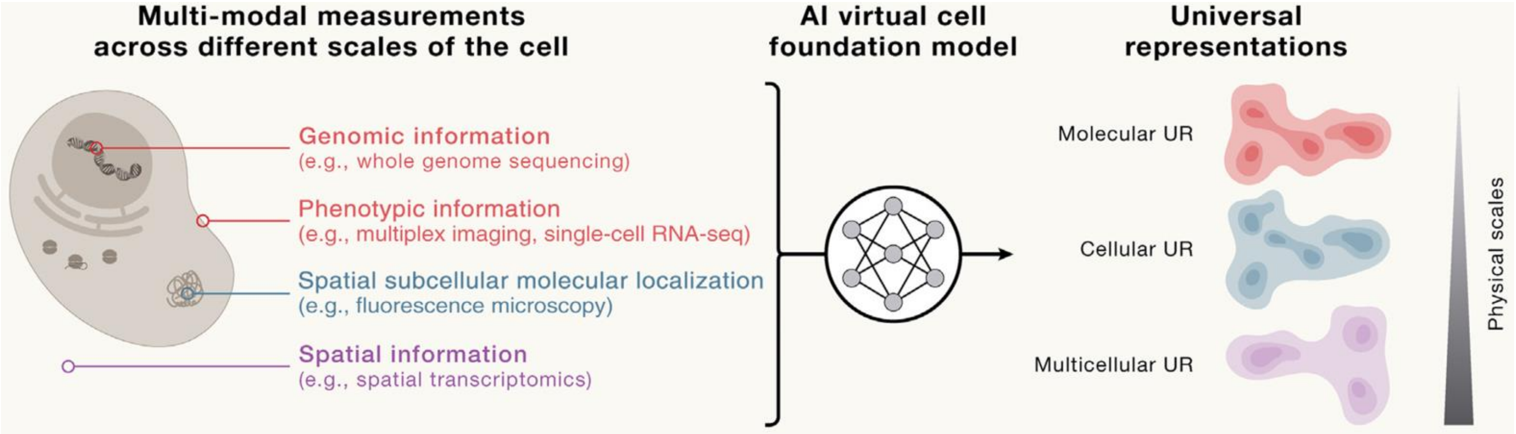
Cell
Leading Edge

CellPress
OPEN ACCESS

Perspective

How to build the virtual cell with artificial intelligence: Priorities and opportunities

Charlotte Bunne,[1,2,3,4,50] Yusuf Roohani,[1,3,5,50] Yanay Rosen,[1,3,50] Ankit Gupta,[3,6] Xikun Zhang,[1,3,7] Marcel Roed,[1,3] Theo Alexandrov,[8,9] Mohammed AlQuraishi,[9] Patricia Brennan,[3] Daniel B. Burkhardt,[11] Andrea Califano,[10,12,13] Jonah Cool,[3] Abby F. Dernburg,[14] Kirsty Ewing,[3] Emily B. Fox,[1,15,16] Matthias Haury,[17] Amy E. Herr,[16,18] Eric Horvitz,[19] Patrick D. Hsu,[5,18,20] Viren Jain,[21] Gregory R. Johnson,[22] Thomas Kalil,[23] David R. Kelley,[24] Shana O. Kelley,[25,26] Anna Kreshuk,[27] Tim Mitchison,[28] Stephani Otte,[17] Jay Shendure,[29,30,31,32] Nicholas J. Sofroniew,[33] Fabian Theis,[34,35,36] Christina V. Theodoris,[37,38] Srigokul Upadhyayula,[14,16,39] Marc Valer,[3] Bo Wang,[40,41] Eric Xing,[42,43] Serena Yeung-Levy,[1,44] Marinka Zitnik,[45,46,47] Theofanis Karaletsos,[3,*] Aviv Regev,[2,*] Emma Lundberg,[3,6,7,48,*] Jure Leskovec,[1,3,*] and Stephen R. Quake[3,7,49,*]



Fundamental difference:

- Whole-cell modeling is knowledge (or rule)-driven
- AI virtual cell is mainly data-driven

立志成才 报国裕民

Challenges to rule-based cell modeling

上海科技大学 ShanghaiTech University

- Existing rule-based cell models often combine assumptions about the underlying biological mechanisms with parameters fit from data, and use explicitly defined mathematical or computational approaches, e.g.:
  - Differential equations
  - Stochastic simulations
  - Agent-based models
- Challenges to the rule-based cell models:
  - Multi-scale modeling: Cells operate on multiple scales across both time and space
  - Diverse processes with massive numbers of interacting components: E.g. gene regulation, metabolic pathways, and signal transduction
  - Nonlinear dynamics: small changes in inputs can lead to complex changes in outputs

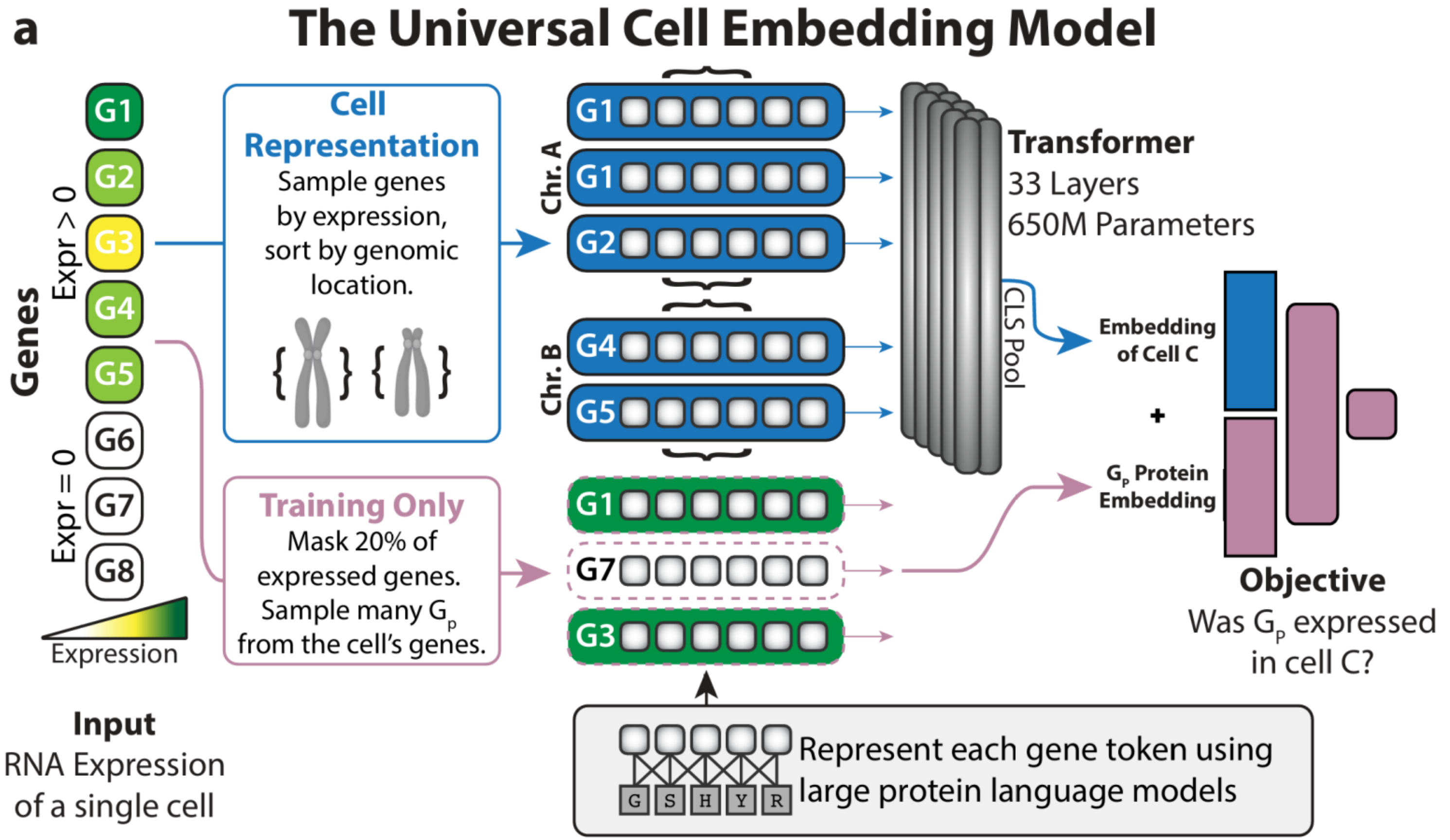
立志成才报国裕民

Two revolutions enable data-driven cell modeling

- **Revolution 1 (experimental): Omics data**
  - Increase in the high-throughput measurement technologies has led to the collection of large and growing datasets
  - Data doubling every 6 months for the past several years
  - The ability to couple the measurements with systematic perturbations
- **Revolution 2 (computational): AI technologies**
  - Enhanced ability to learn patterns and processes directly from data
  - Without needing explicit rules or human annotation
  - E.g. AlphaFold and protein language models
  - Predictive, generative, and queryable
- **An ambitious vision of AIVC**: A multi-scale, multi-modal, large-neural-network-based, fully data-driven model
  - Enable fast-paced in silico studies
  - Bridges between computational methods and confirmatory wet-lab experiments

Universal representation (UR)

- **Definition**: Universal Representation (UR) is a multi-scale, multi-modal embedding that captures the state and behavior of biological entities across different contexts and modalities.
- **Purpose**: To create a unified framework for representing and analyzing biological data, enabling high-fidelity simulations and hypothesis generation.

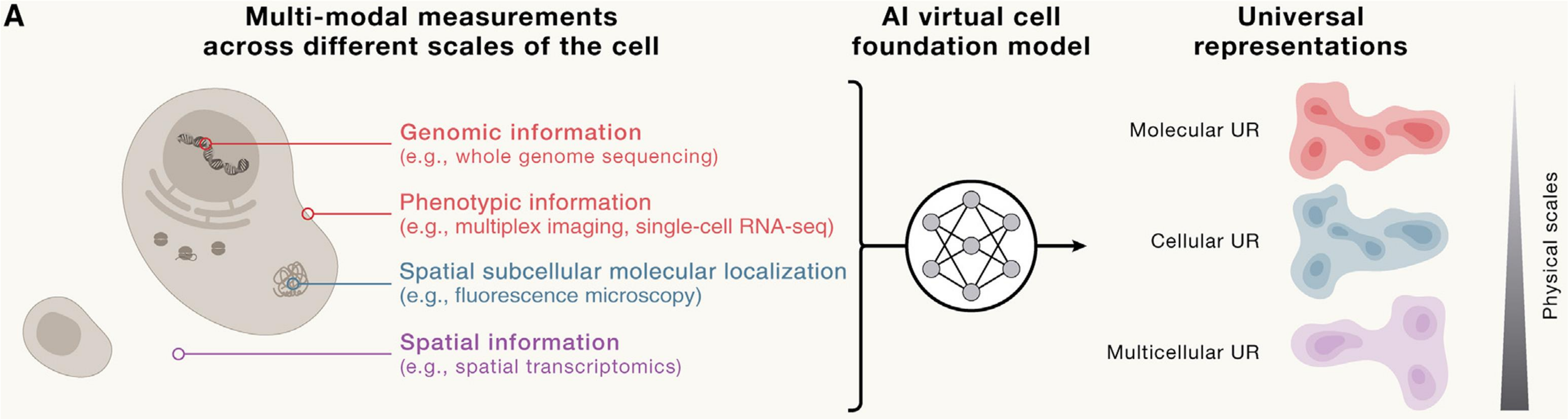


Rosen, Y. et al. (2023) Universal cell embeddings: A foundation model for cell biology. bioRxiv.

Features of universal representation (UR)

上海科技大学 ShanghaiTech University

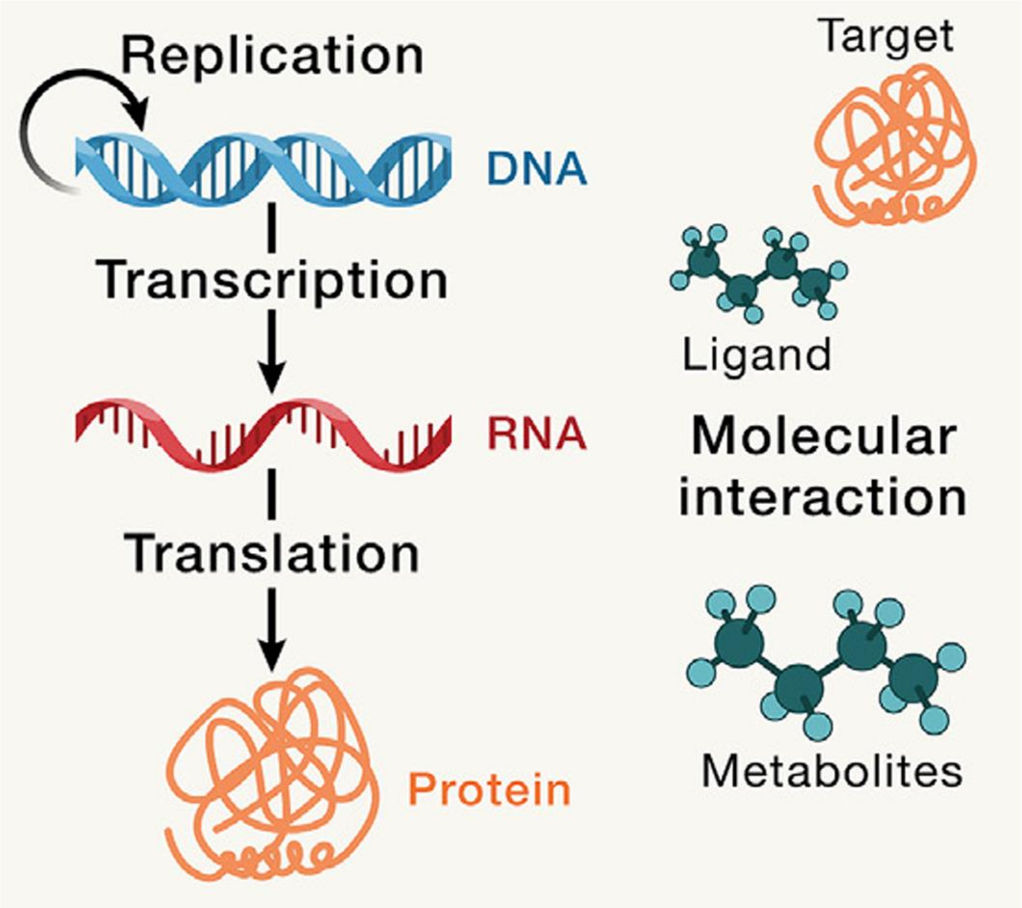
- **Multi-Scale Integration**: UR integrates data across molecular, cellular, and multicellular levels
  - **Molecular Scale**: Represents individual molecules (DNA, RNA, proteins) and their interactions
  - **Cellular Scale**: Captures the state of individual cells, including gene expression, protein levels, and spatial organization
  - **Multicellular Scale**: Models interactions between cells and their organization into tissues and organs
- **Multi-Modal Data**: UR can be generated from various data types, including sequencing, imaging, and proteomics

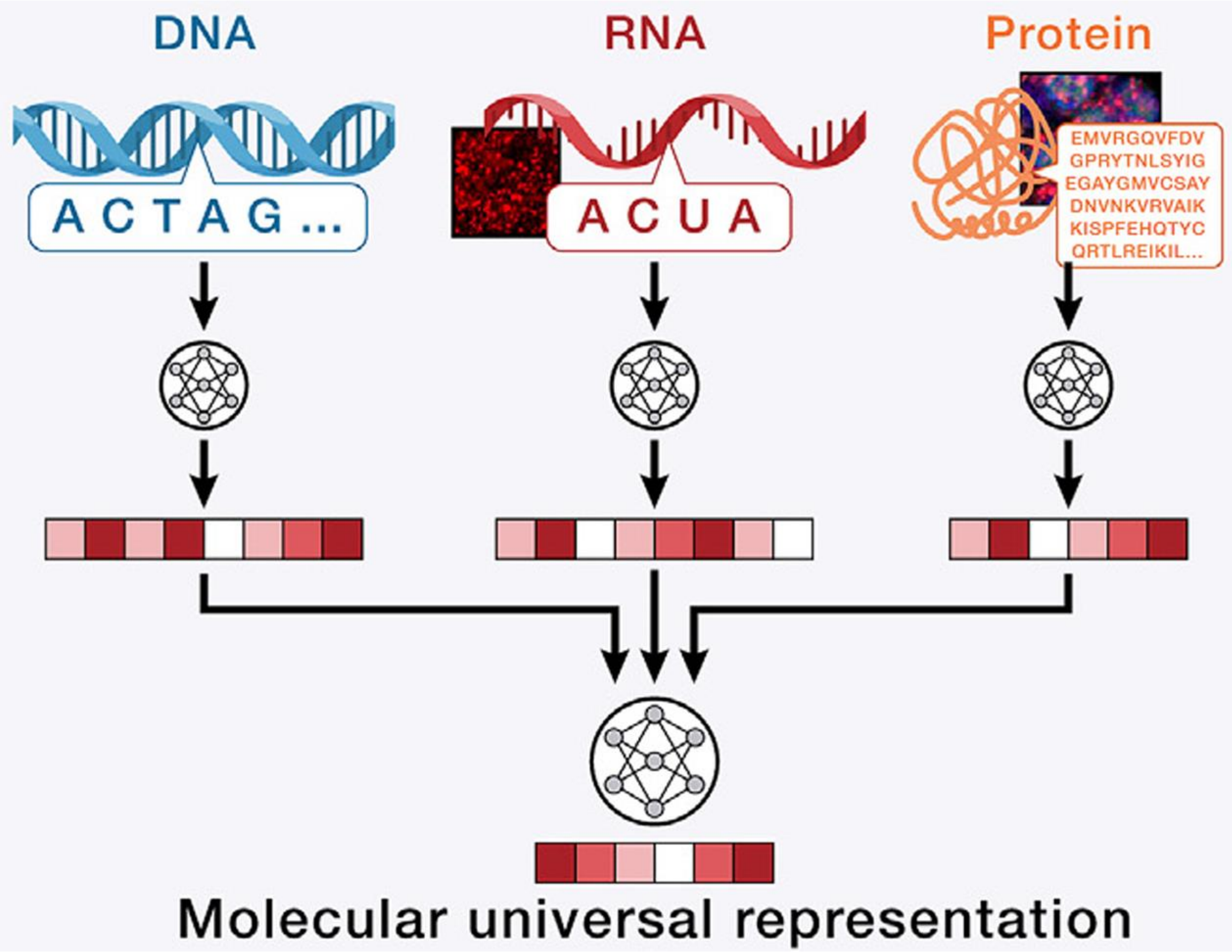


C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

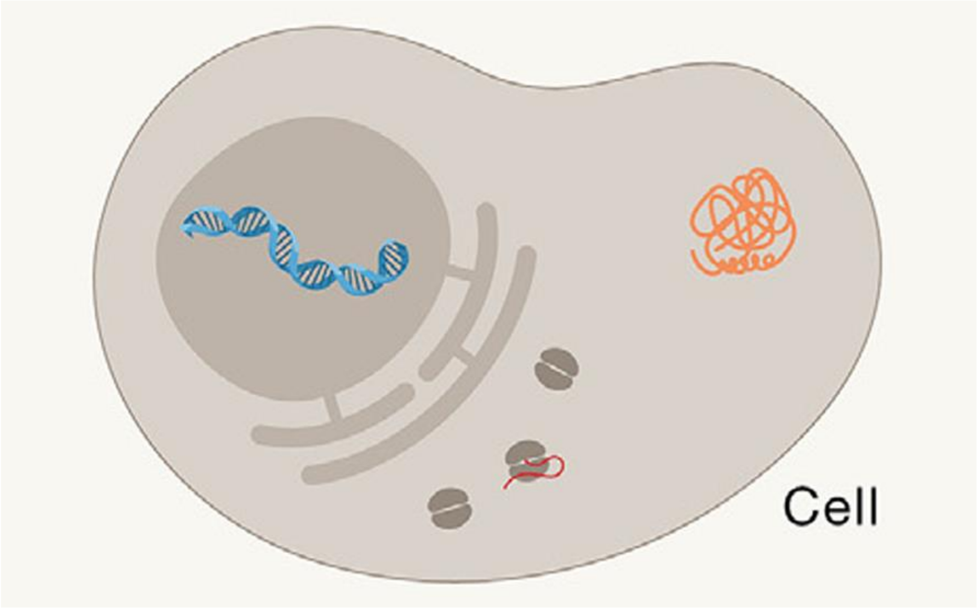
立志成才报国裕民

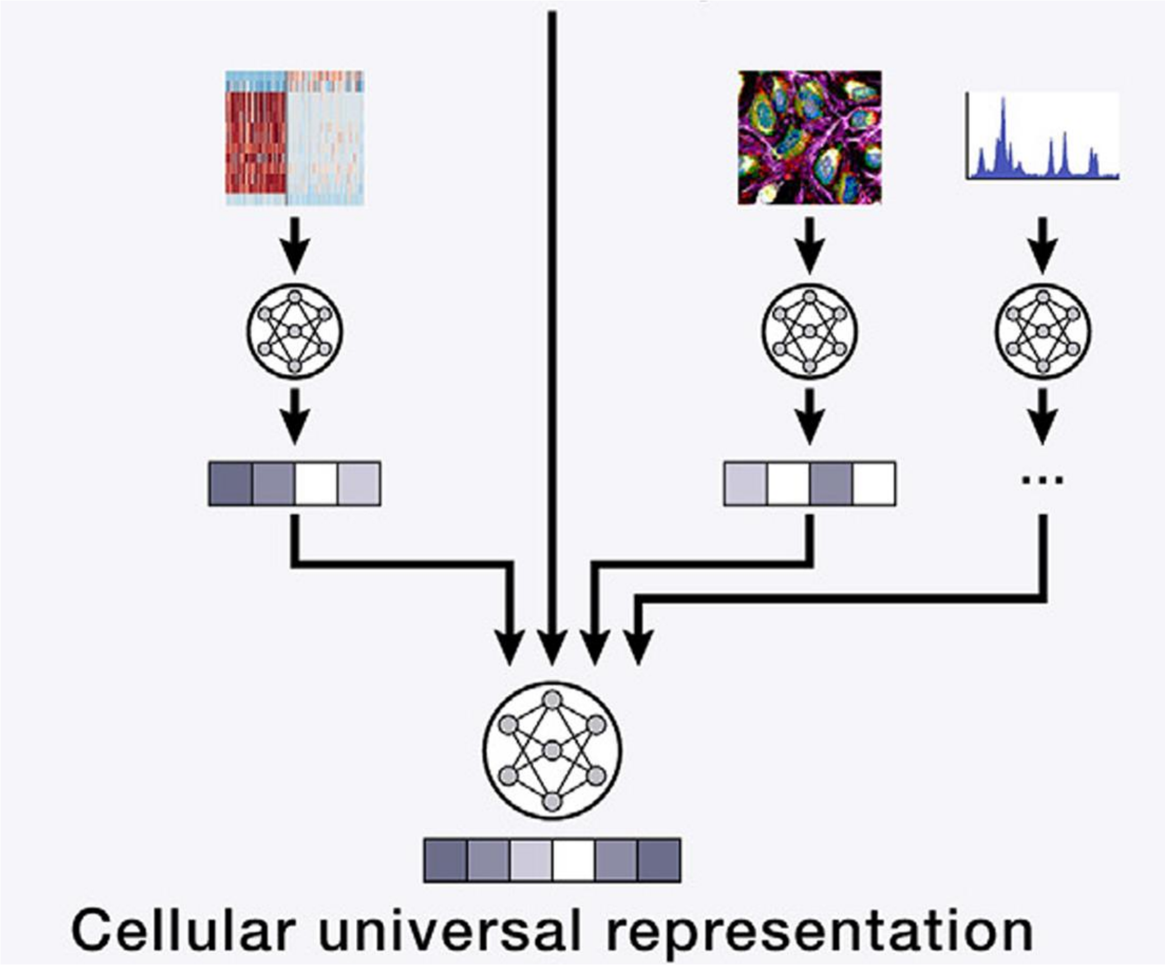
UR at molecular scale





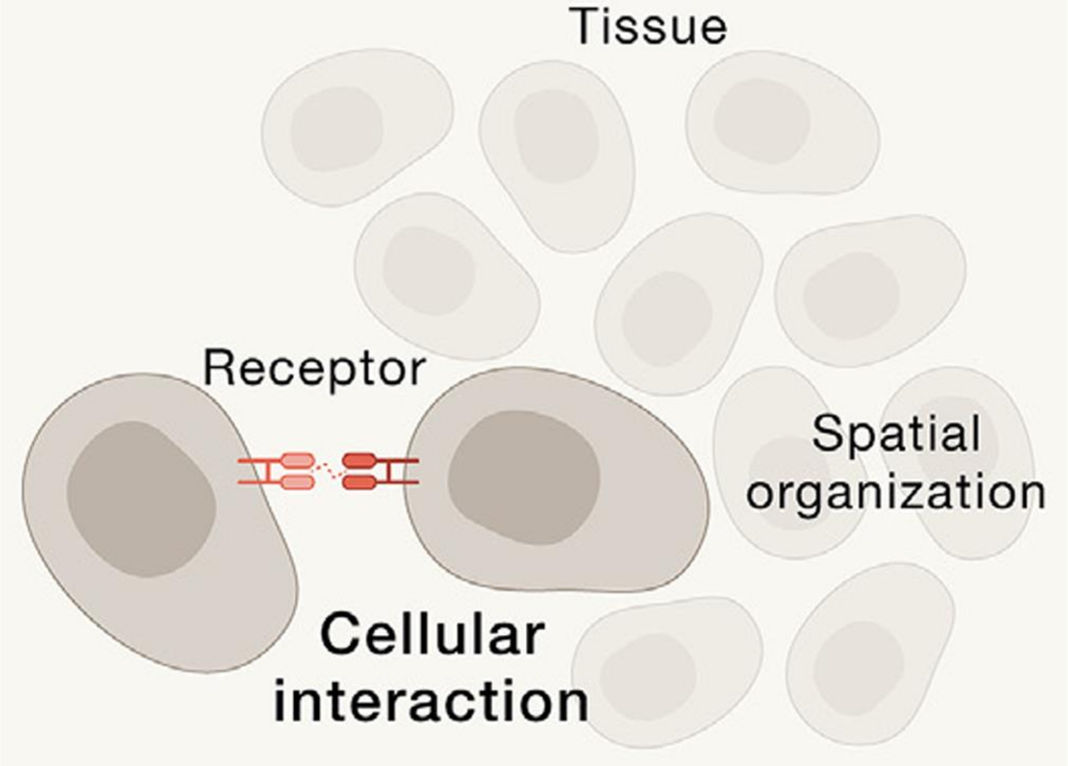
UR at cellular scale

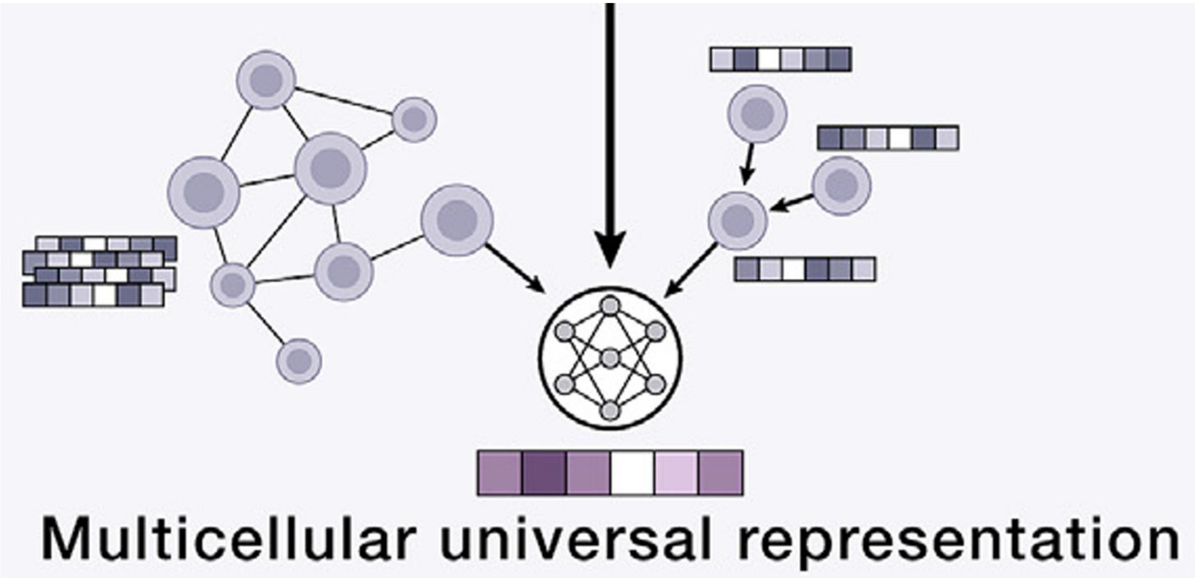




立志成才报国裕民

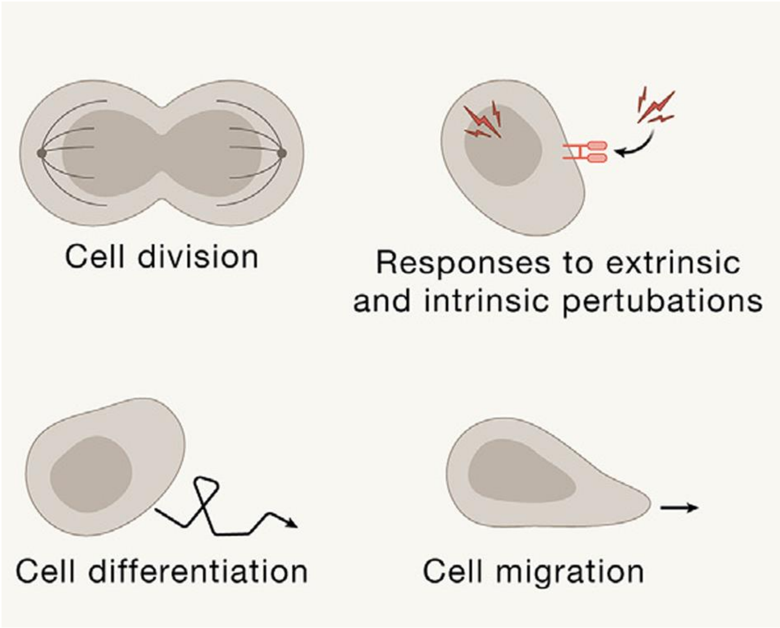
UR at multicellular scale

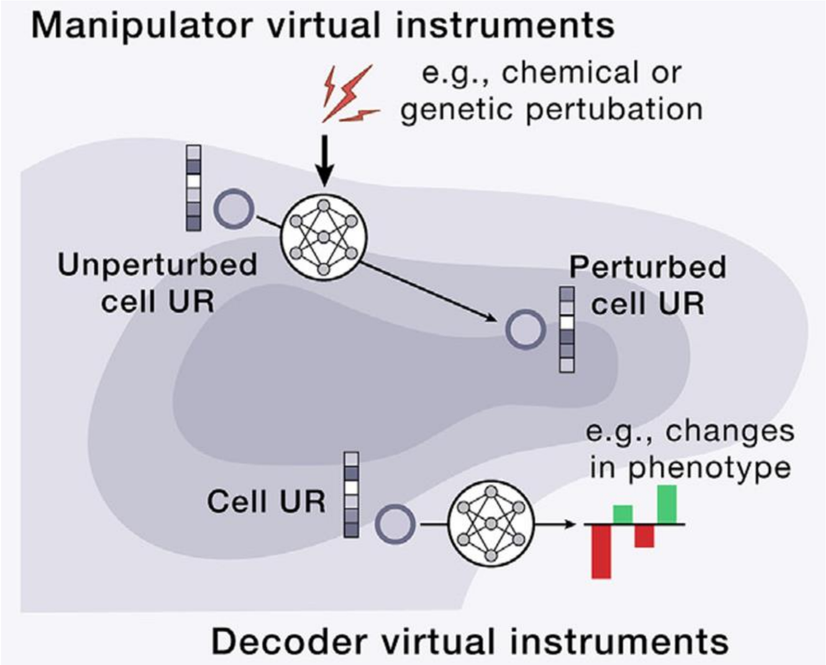




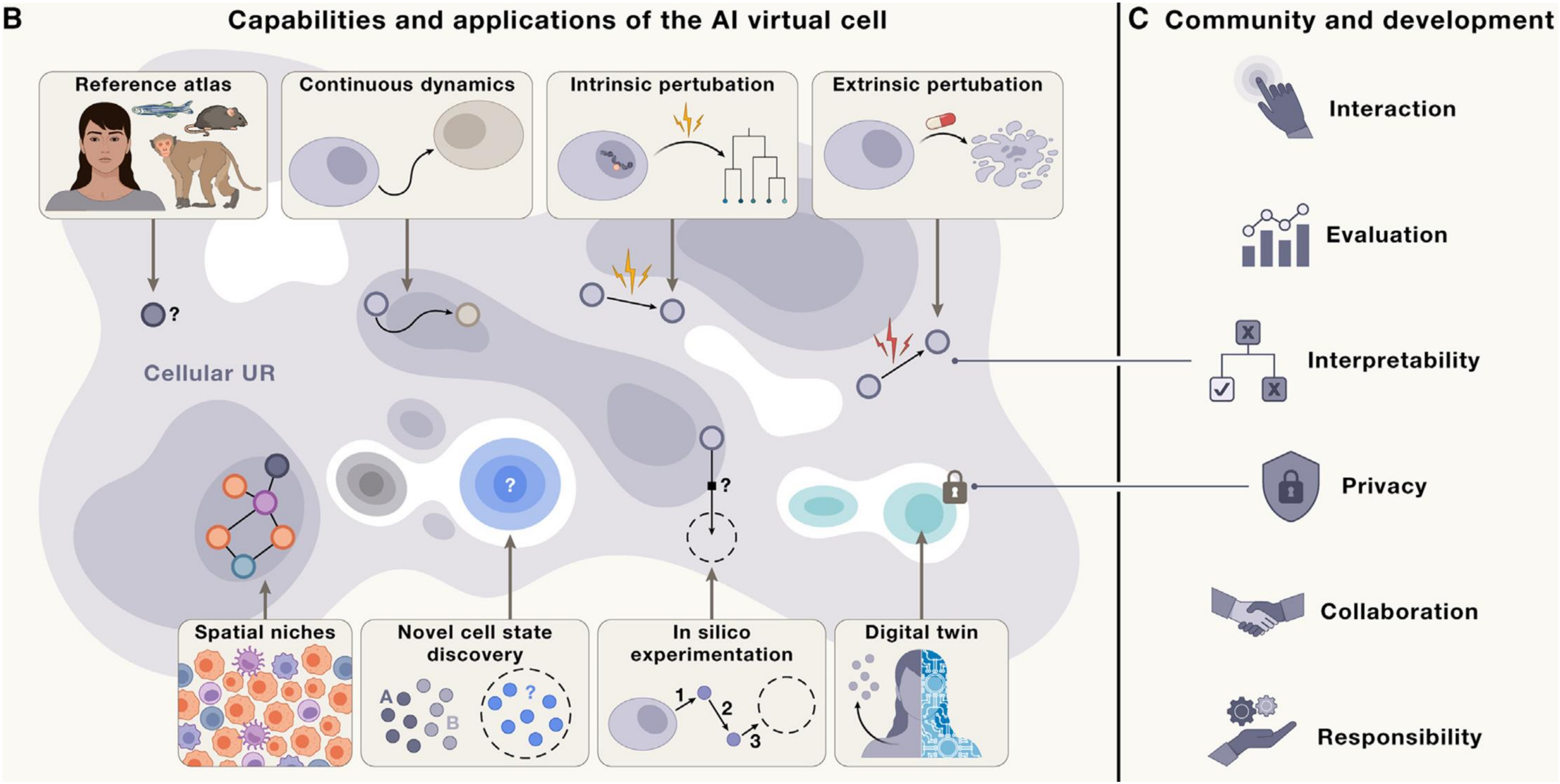
Virtual instruments (VIs)

- **Virtual instruments (VIs)** are the tools that operate on UR embeddings and perform various functions and tasks
  - **Manipulators**: By altering molecular, cellular and tissue URs, manipulator VIs abstract complex dynamic processes as the transitions between (distributions of) their representations
  - **Decoders**: Take an embedding of biological entities and predict one or more concrete properties, e.g. physical structure, cell type/state, fitness, expression, or drug response



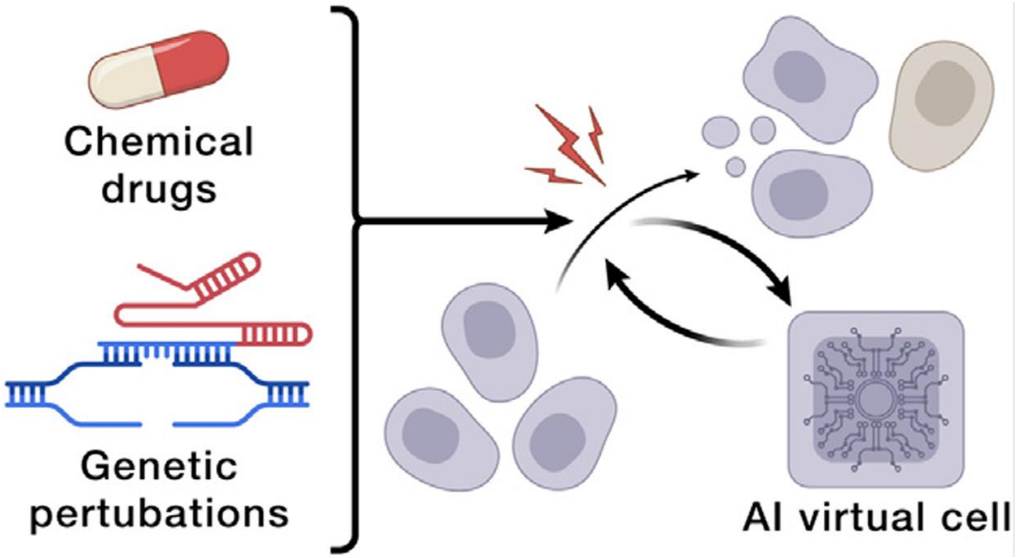


Capabilities and applications of AIVC

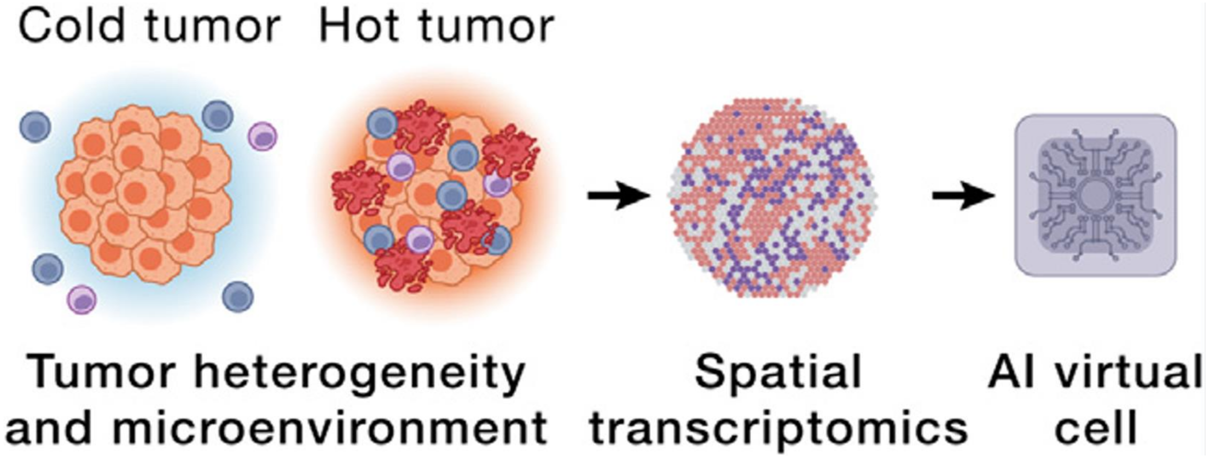


C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

Applications of AIVC



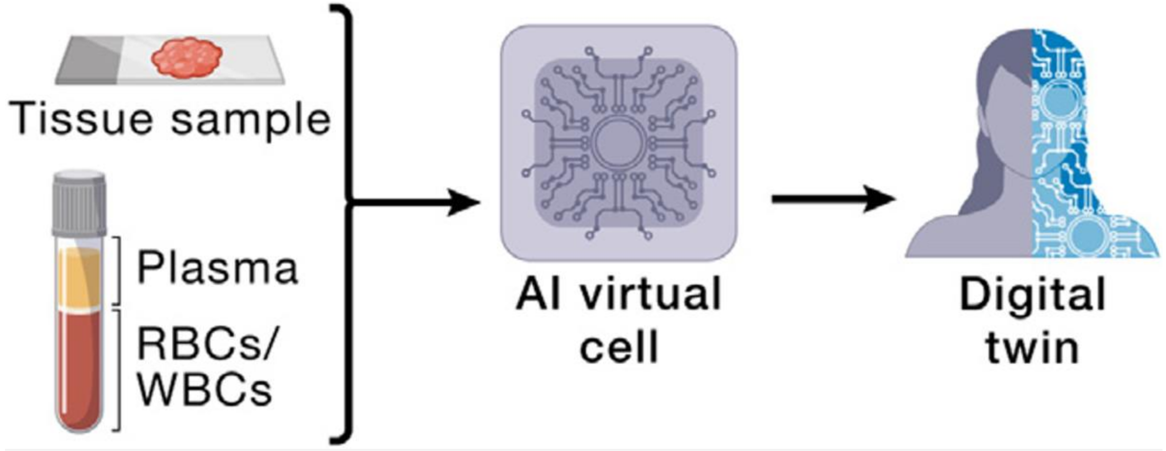
Cell engineering to enable phenotypic drug discovery and cell-based therapeutics



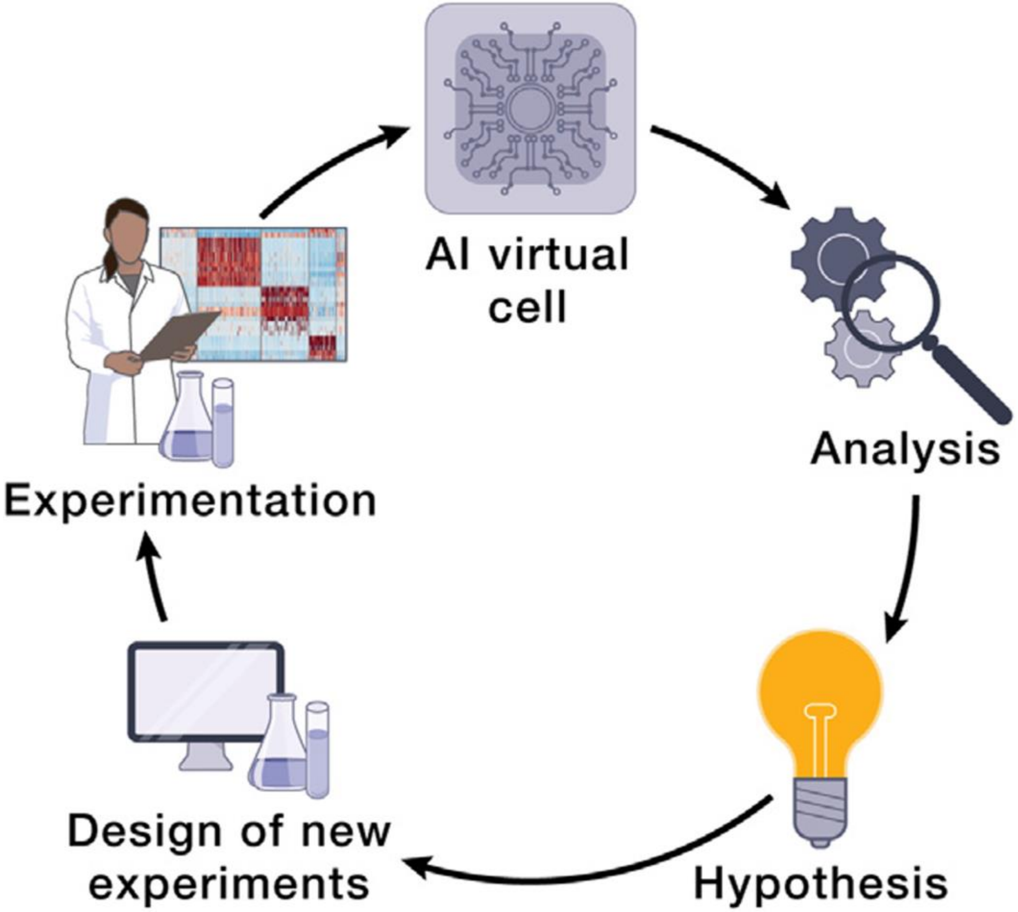
Unlocking the power of spatial biology to fight cancer

C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

Applications of AIVC



Diagnostic virtual cell models for individual patients



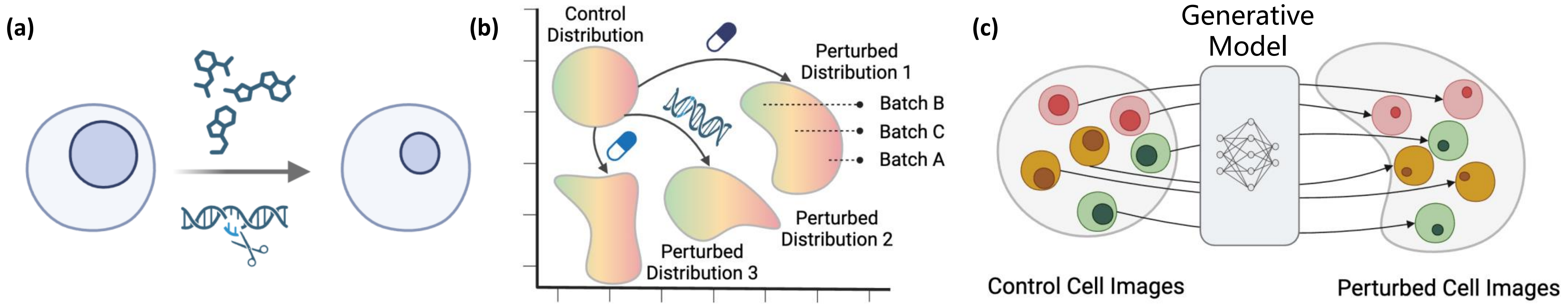
A hypothesis-generating framework for scientific research

C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

Generative AI for perturbation modeling

基因扰动和细胞表型变化的关联预测

基于生成式人工智能模拟扰动引起的细胞表型变化



Zhang Y, Su Y, Wang C, et al. CellFlow: Simulating Cellular Morphology Changes via Flow Matching. arXiv, 2025.

Challenges of AIVC

- Data needs and requirement
- Model evaluation
- Interpretability and interaction
- Open collaboration as a scientific community

Recap

- In this lecture, we have learned:
  - The basics of whole-cell modeling (aims, methodology, and challenges)
  - A classic ODE model of cell cycle control
  - AI virtual cell (AIVC) modeling learns directly from data (i.e. it is data-driven)
  - Universal representations (URs) at molecular, cellular and multicellular (e.g. tissue and organ) scales are the foundation of AIVC models
  - Virtual instruments (VIs) enable prediction of cell phenotypic changes upon perturbations and explanation of mechanisms
  - Despite an optimistic future, AIVC is faced with several challenges

References

- Goldberg et al. (2018), "Emerging whole-cell modeling principles and methods", Current Opinion in Biotechnology, **51**:97–102.
- Tyson, Chen, and Novak (2001), "Network Dynamics and Cell Physiology", Nature Reviews Molecular Cell Biology, **2**:908-916, 2001.
- Novak and Pataki (2001), "Mathematical model of the cell division cycle of fission yeast", Chaos **11**(1):277-286.
- C. Bunne, et al. (2024) How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 187, pp. 7045 – 7063.
- Rosen, Y. et al. (2023) Universal cell embeddings: A foundation model for cell biology. bioRxiv.
- Zhang Y, Su Y, Wang C, et al. (2025) CellFlow: Simulating Cellular Morphology Changes via Flow Matching. arXiv.